MAXIMUM TWO-DISTANCE SETS IN EUCLIDEAN SPACES OF DIMENSION NINE, TEN, ELEVEN AND TWELVE

BRYAN ZACHARY BU, KARAM GILL, HYUNJUN YI

ABSTRACT. A two-distance set is a finite set of points in $\mathbb{R}^d$ whose pairwise distances take only two values. Let $g(d)$ be the largest cardinality of a two-distance set in $\mathbb{R}^d$. The values $g(d)$ for $d \leq 8$ have been known since Lisoněk's work of 1997; the case $d = 9$, where $45 \leq g(9) \leq 55$, has remained open since then. We prove that $g(9) = 45$ and that, up to similarity, there are exactly two 45-point two-distance sets in $\mathbb{R}^9$: the Johnson set $J(10, 2)$ and Lisoněk's 45-point set in $\mathbb{R}^8$. We also prove $g(10) = 55$, $g(11) = 66$, $g(12) = 78$, each attained only by $J(d + 1, 2)$, and that Lisoněk's set is the unique 45-point two-distance set in $\mathbb{R}^8$. The proofs combine a structural reduction with an exhaustive, symmetry-reduced computer search: the Larman–Rogers–Seidel theorem forces the distance ratio $\sqrt{2}$, after which the short differences of the configuration generate an ADE root lattice, and every configuration with connected short-distance graph is a clique in a finite graph built from the vectors of norm 2 and 4 of a root lattice of rank $\leq d$. Disconnected configurations are handled by a separate geometric lemma. The search reproduces every previously known value $g(d)$, $d \leq 8$, and the new negative results were cross-checked with an independent SAT-based implementation.

1. INTRODUCTION

A set $X \subset \mathbb{R}^d$ is a *two-distance set* if $\{ \|x - y\| : x \neq y \in X \}$ has exactly two elements. Let $g(d)$ denote the maximum cardinality of a two-distance set in $\mathbb{R}^d$. The problem of determining $g(d)$ goes back to Kelly [10] and Croft [6]; the general bounds

$$\binom{d+1}{2} \;\leq\; g(d) \;\leq\; \binom{d+2}{2}$$

are due to the midpoint construction (the set $J(d + 1, 2)$ of midpoints of the edges of a regular simplex) and to Blokhuis [3], improving Bannai–Bannai–Stanton [1]. Einhorn and Schoenberg [8] settled $d \leq 3$, and Lisoněk [12] classified all maximal two-distance sets for $d \leq 7$ by computer and found a 45-point set in $\mathbb{R}^8$ which attains Blokhuis' bound; hence

$$g(1), \ldots, g(8) \;=\; 3,\; 5,\; 6,\; 10,\; 16,\; 27,\; 29,\; 45.$$

For $d = 9$ only $45 \leq g(9) \leq 55$ was known, and the problem is listed as open in several recent papers [4, 2, 15]; Bannai, Bannai, Lee, Xiang and Yu [2] also note that the classification of the maximum sets in $\mathbb{R}^8$ is open.

Theorem 1.1. $g(9) = 45$. *Up to similarity there are exactly two* 45*-point two-distance sets in* $\mathbb{R}^9$*: the Johnson set* $J(10, 2)$ *(which spans* $\mathbb{R}^9$ *and lies on a sphere), and Lisoněk's set* L_8 *(which spans an* 8*-dimensional subspace and is not spherical).*

Theorem 1.2. $g(10) = 55$, $g(11) = 66$ *and* $g(12) = 78$. *For* $d \in \{10, 11, 12\}$ *the set* $J(d + 1, 2)$ *is, up to similarity, the unique two-distance set of cardinality* $g(d)$ *in* $\mathbb{R}^d$.

Theorem 1.3. *Lisoněk's set* L_8 *is, up to similarity, the unique* 45*-point two-distance set in* $\mathbb{R}^8$.

The method also yields a structural statement which we find of independent interest.

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Theorem 1.4. *Let $9 \le d \le 12$ and let $X \subset \mathbb{R}^d$ be a two-distance set with $|X| > 2d+1$, scaled so that its squared distances are 2 and 4. Let $\Gamma(X)$ be the graph on X joining points at squared distance 2. Then every connected component X_i of $\Gamma(X)$ satisfies $X_i - X_i \subset R_i$ for an ADE root lattice R_i of rank at most $\dim \mathrm{aff}(X_i)$; the spans of the R_i are mutually orthogonal; and if $\Gamma(X)$ is disconnected, every component is spherical. In particular, if $\Gamma(X)$ is connected then $X \subset p + R$ for a root lattice $R \subset \mathbb{R}^d$ and any $p \in X$.*

The proofs have two parts. The first is a short reduction (Sections 3 and 4), whose only non-trivial inputs are the Larman–Rogers–Seidel theorem, the classification of even lattices generated by norm-2 vectors, and an elementary geometric lemma about disconnected configurations. It shows that for $9 \le d \le 12$ the determination of $g(d)$ and of the extremal configurations is equivalent to computing the clique numbers of finitely many explicit graphs $G(R)$ (Definition 2.2), one for each root lattice R of rank d, together with an enumeration of the near-maximum cliques of a handful of them. The second part is the computation (Section 5): there are 57 root lattices of rank 9 (88, 128 and 193 of ranks 10, 11, 12), and the largest graphs have up to 8208 vertices. A branch-and-bound clique solver with symmetry reduction by the Weyl group solves all of them; the total time is under two minutes. Section 7 describes how the results were validated: the pipeline reproduces every known value $g(d)$, $d \le 8$, and the uniqueness of the extremal sets in dimensions 6, 7 and 8; the negative results for rank 9 were confirmed by a second, independently written solver (SAT-based for the one lattice where a plain, symmetry-free search does not terminate quickly). All code and data are provided with the paper.

2. Preliminaries

Throughout, $\langle \cdot, \cdot \rangle$ is the standard inner product and $\|x\|^2 = \langle x, x \rangle$.

Theorem 2.1 (Larman–Rogers–Seidel [11], Neumaier [13])**.** *Let $X \subset \mathbb{R}^d$ be a two-distance set with distances $a < b$ and $|X| > 2d+1$. Then $a^2/b^2 = (k-1)/k$ for an integer k with $2 \le k \le \frac{1}{2} + \sqrt{d/2}$.*

For $d \le 12$ we have $\frac{1}{2} + \sqrt{d/2} < 3$, so $k = 2$ and $b^2 = 2a^2$. *Hence every two-distance set in $\mathbb{R}^d$, $d \le 12$, with more than $2d+1$ points is similar to one with squared distances 2 and 4.* All two-distance sets below are assumed scaled in this way.

For such a set X let $\Gamma(X)$ be the graph with vertex set X in which $x \sim y$ iff $\|x-y\|^2 = 2$; we call it the *short-distance graph*. A set X is *spherical* if it lies on a sphere; its *circumradius* $\rho(X)$ is the radius of the smallest sphere in $\mathrm{aff}(X)$ containing X (if X is spherical, this sphere is unique and its centre lies in $\mathrm{aff}(X)$).

Root lattices. An *even lattice* is a free abelian group $\Lambda \subset \mathbb{R}^n$ of finite rank, discrete in $\mathbb{R}^n$, on which $\|\cdot\|^2$ takes even integer values (so that $\langle \cdot, \cdot \rangle$ is integer-valued on Λ). A *root lattice* is an even lattice generated by its vectors of norm $\|v\|^2 = 2$ (its *roots*). By Witt's classification (see e.g. [7, Ch. 1] or [5, Ch. 4]), every root lattice is an orthogonal direct sum of copies of the irreducible root lattices A_n ($n \ge 1$), D_n ($n \ge 4$), E_6, E_7, E_8, uniquely up to order. The number of root lattices of rank exactly r is the number of ways of writing r as a sum of ranks of irreducible types; for $r = 8, 9, 10, 11, 12$ it is $39, 57, 88, 128, 193$. We use the standard coordinates

$$A_n = \{x \in \mathbb{Z}^{n+1} : \textstyle\sum x_i = 0\}, \qquad D_n = \{x \in \mathbb{Z}^n : \textstyle\sum x_i \equiv 0 \ (2)\}, \qquad E_8 = D_8 \cup \big(D_8 + \tfrac{1}{2}(1, \dots, 1)\big),$$

and E_7, E_6 are the sublattices of E_8 orthogonal to a root, respectively to an A_2-sublattice.

For a root lattice R write $R_m = \{v \in R : \|v\|^2 = m\}$. The vectors of norm 4 are: in A_n, the vectors $e_i + e_j - e_k - e_l$ with i, j, k, l distinct; in D_n, the vectors $\pm 2e_i$ and $\pm e_i \pm e_j \pm e_k \pm e_l$; in E_8, the 2160 vectors of the second shell; and in a direct sum $R' \oplus R''$, the norm-4 vectors of the summands together with all sums $r' + r''$ of a root of R' and a root of R''. (These

lists follow at once from the coordinates; the counts $|R_2|, |R_4|$ for $A_3, A_4, D_4, D_5, E_6, E_7, E_8$ are $(12,6), (20,30), (24,24), (40,90), (72,270), (126,756), (240,2160)$, in agreement with the theta series.)

The Weyl group $W(R)$ is generated by the reflections $s_r(x) = x - \langle x, r\rangle r$, $r \in R_2$; each s_r preserves R, R_2, R_4 and all distances.

Definition 2.2. For a root lattice R let $G(R)$ be the graph with vertex set $R_2 \cup R_4$ in which $v \sim w$ iff $\|v-w\|^2 \in \{2,4\}$. Let $\omega(R)$ be its clique number, and for $a \geq 1$ let

$$f(a) = \max\{\, 1 + \omega(R) : R \text{ a root lattice of rank} \leq a \,\}, \qquad f(0) = 1.$$

If R has rank $< a$ then $R \subset R \oplus A_1$ and $G(R)$ is an induced subgraph of $G(R \oplus A_1)$, so $\omega(R) \leq \omega(R \oplus A_1)$; hence the maximum in $f(a)$ may be taken over root lattices of rank exactly a.

3. The lattice reduction

Lemma 3.1. *Let $X \subset \mathbb{R}^d$ be a two-distance set with squared distances 2 and 4. Then $\langle x-y, z-w\rangle \in \mathbb{Z}$ for all $x, y, z, w \in X$, and the additive group $\Lambda(X)$ generated by $X - X$ is an even lattice of rank at most d.*

Proof. For $x, y, z \in X$ the polarisation identity gives $\langle x-z, y-z\rangle = \frac{1}{2}(\|x-z\|^2 + \|y-z\|^2 - \|x-y\|^2)$, which is an integer because each of the three squared norms lies in $\{0,2,4\}$ and their sum is even. Then $\langle x-y, z-w\rangle = \langle x-y, z-y\rangle - \langle x-y, w-y\rangle \in \mathbb{Z}$. So $\Lambda = \Lambda(X)$ is generated by finitely many vectors of even norm with integral pairwise inner products; hence $\|\lambda\|^2 \in 2\mathbb{Z}$ for all $\lambda \in \Lambda$. In particular $\|\lambda\|^2 \geq 2$ for $0 \neq \lambda \in \Lambda$, so Λ is a discrete subgroup of $\mathbb{R}^d$, i.e. a lattice, and its rank is at most d. □

Lemma 3.2. *In the situation of Lemma 3.1, let $R(X) \subset \Lambda(X)$ be the subgroup generated by the differences $x - y$ with $\|x-y\|^2 = 2$, i.e. by the edges of $\Gamma(X)$. Then $R(X)$ is a root lattice of rank at most d.*

Proof. $R(X)$ is a sublattice of the even lattice $\Lambda(X)$, hence an even lattice, and it is generated by vectors of norm 2; so it is a root lattice by definition, and Witt's classification applies. □

Lemma 3.3 (Connected configurations are cliques)**.** *Let $X \subset \mathbb{R}^d$ be a two-distance set with squared distances 2 and 4 whose short-distance graph $\Gamma(X)$ is connected, and let $p \in X$. Then $X - p \subset R(X)$, and $C := (X-p) \setminus \{0\}$ is a clique of $G(R(X))$ of size $|X| - 1$. Conversely, for every root lattice R and every clique C of $G(R)$, the set $\{0\} \cup C$ is a set with squared distances in $\{2,4\}$. Consequently*

$$|X| \leq f(d) \qquad \textit{whenever } \Gamma(X) \textit{ is connected and } X \subset \mathbb{R}^d.$$

Proof. For $x \in X$ choose a path $p = x_0, x_1, \ldots, x_m = x$ in $\Gamma(X)$; then $x - p = \sum (x_i - x_{i-1})$ is a sum of edge vectors, so $x - p \in R(X)$. The vectors $x - p$, $x \neq p$, have norm $\|x-p\|^2 \in \{2,4\}$ and pairwise satisfy $\|(x-p)-(y-p)\|^2 = \|x-y\|^2 \in \{2,4\}$, so they form a clique of $G(R(X))$. The converse is immediate from the definition of $G(R)$: all vertices have norm 2 or 4 (distance to 0) and adjacent vertices are at squared distance 2 or 4. The bound follows since $\operatorname{rank} R(X) \leq d$. □

Note that $\{0\} \cup C$ is a genuine two-distance set unless all of its distances coincide, which for $|C| \geq d+2$ is impossible; so for our purposes $f(d)$ is exactly the maximum size of a connected configuration in $\mathbb{R}^d$ (once $f(d) \geq d+3$, which holds for $d \geq 3$).

4. Disconnected configurations

Lemma 4.1. *Let $X \subset \mathbb{R}^d$ be a two-distance set with squared distances 2 and 4 whose short-distance graph has $m \geq 2$ connected components $X_1, \dots, X_m$. Put $U_i = \operatorname{span}(X_i - X_i)$ and $a_i = \dim U_i$. Then:*

(i) *$U_i \perp U_j$ for $i \neq j$.*
(ii) *Each X_i is spherical: there are $c_i \in \operatorname{aff}(X_i)$ and $\rho_i \geq 0$ with $\|x - c_i\|^2 = \rho_i^2$ for all $x \in X_i$.*
(iii) *$c_i - c_j \perp U_k$ for all i, j, k, and $\|c_i - c_j\|^2 + \rho_i^2 + \rho_j^2 = 4$ for all $i \neq j$.*
(iv) *$\dim \operatorname{aff}(X) = \sum_i a_i + \dim \operatorname{span}\{c_i - c_1 : 1 \leq i \leq m\}$. In particular $\dim \operatorname{aff}(X) \geq \sum_i a_i$, with equality if and only if all c_i coincide, and $\dim \operatorname{aff}(X) \geq \sum_i a_i + 1$ if at least two of the X_i are single points.*
(v) *$|X_i| \leq f(a_i)$ for every i.*

Proof. Points in different components are at squared distance 4. (i) For $x, x' \in X_i$ and $y, y' \in X_j$, $i \neq j$, expanding $\|x - y\|^2 = \|x - y'\|^2 = \|x' - y\|^2 = \|x' - y'\|^2 = 4$ gives $\langle x - x', y - y' \rangle = \frac{1}{2}(\|x - y'\|^2 + \|x' - y\|^2 - \|x - y\|^2 - \|x' - y'\|^2) = 0$.

(ii) Fix $j \neq i$ and $y \in X_j$. Every $x \in X_i$ satisfies $\|x - y\|^2 = 4$, so X_i lies on the sphere of radius 2 about y. Intersecting with $\operatorname{aff}(X_i)$ shows that X_i lies on the sphere in $\operatorname{aff}(X_i)$ with centre c_i, the orthogonal projection of y to $\operatorname{aff}(X_i)$, and radius $\rho_i = \sqrt{4 - \operatorname{dist}(y, \operatorname{aff} X_i)^2}$. Since X_i spans $\operatorname{aff}(X_i)$, a sphere in $\operatorname{aff}(X_i)$ containing X_i is unique, so c_i and ρ_i do not depend on y (or j). Moreover $y - c_i \perp U_i$ for every $y \in X_j$, $j \neq i$.

(iii) Let $i \neq j$ and $y \in X_j$. We have $c_i - c_j = (c_i - y) + (y - c_j)$ with $c_i - y \perp U_i$ (by (ii)) and $y - c_j \in U_j \perp U_i$ (by (i), as $c_j \in \operatorname{aff} X_j$). Hence $c_i - c_j \perp U_i$, and by symmetry $c_i - c_j \perp U_j$. For $k \notin \{i, j\}$, $c_i - c_j = (c_i - c_k) - (c_j - c_k)$ is orthogonal to U_k by what was just shown. Finally, for $x \in X_i$, $y \in X_j$, write $x - y = (x - c_i) + (c_i - c_j) + (c_j - y)$; the three summands lie in U_i, in $(U_i \oplus U_j)^\perp$ and in U_j respectively, so they are mutually orthogonal and $4 = \|x - y\|^2 = \rho_i^2 + \|c_i - c_j\|^2 + \rho_j^2$.

(iv) $\operatorname{aff}(X) - c_1$ is spanned by $\bigcup_i (X_i - c_i) \cup \{c_i - c_1\}$, i.e. by $U_1 \oplus \cdots \oplus U_m$ together with $\operatorname{span}\{c_i - c_1\}$, and by (i) and (iii) these are orthogonal. If $X_i = \{p\}$ and $X_j = \{q\}$ are single points then $c_i = p \neq q = c_j$, so $\operatorname{span}\{c_i - c_1\} \neq 0$.

(v) If $a_i = 0$ then $|X_i| = 1 = f(0)$. Otherwise X_i is a two-distance set (or a set with a single distance $\sqrt{2}$, which is covered by the same argument) in the a_i-dimensional space $\operatorname{aff}(X_i)$ whose short-distance graph is connected, and Lemma 3.3 gives $|X_i| \leq f(a_i)$. □

The values of f used below are those in Table 1, obtained by computation in Section 5:

$$f(0), \dots, f(12) = 1,\ 2,\ 4,\ 6,\ 10,\ 16,\ 27,\ 29,\ 45,\ 45,\ 55,\ 66,\ 78.$$

Corollary 4.2. *Let $X \subset \mathbb{R}^d$ be a two-distance set with squared distances $2, 4$ and disconnected short-distance graph, with components $X_1, \dots, X_m$ numbered so that $a_1 \geq a_2 \geq \dots$.*

(a) *If $d = 9$ and $|X| \geq 45$, then one of the following holds:*
  (a1) *$m = 2$, $(a_1, a_2) = (9, 0)$: $X_2 = \{p\}$, $c_1 = p$, X_1 lies on a sphere of radius 2 and $|X_1| = |X| - 1$;*
  (a2) *$m = 2$, $(a_1, a_2) = (8, 0)$: X_1 is a spherical $|X| - 1$-point set spanning an 8-space, with $\rho_1^2 \leq 4$;*
  (a3) *$m = 2$, $(a_1, a_2) = (8, 1)$: X_2 is a pair at squared distance 2, and X_1 is spherical, spans an 8-space, has $\rho_1^2 = 7/2$ and $|X_1| = |X| - 2$;*
  (a4) *$m = 3$, $(a_1, a_2, a_3) = (8, 0, 0)$: X_1 is spherical, spans an 8-space, has $\rho_1^2 = 3$ and $|X_1| = |X| - 2$.*
(b) *If $d \in \{10, 11, 12\}$ and $|X| \geq \binom{d+1}{2}$, then $m = 2$, $(a_1, a_2) = (d, 0)$, $X_2 = \{p\}$, $c_1 = p$, X_1 lies on a sphere of radius 2 and $|X_1| = |X| - 1$.*

Proof. By Lemma 4.1(iv),(v), $\sum a_i \le d$ and $|X| \le \sum_i f(a_i)$. A direct enumeration of all multisets $\{a_i\}$ with $m \ge 2$, $\sum a_i \le d$ and $\sum f(a_i) \ge N$, using the values of f above, gives, for $d = 9$, $N = 45$: $(9,0)$, $(9,0,0)$, $(8,1)$, $(8,0)$, $(8,0,0)$, $(8,1,0)$, $(8,1,0,0)$, $(8,0,0,0)$, and the same multisets with further single points appended; for $d \in \{10,11,12\}$, $N = \binom{d+1}{2}$: $(d,0)$, $(d,0,0)$, and so on. (The next largest sums are $f(9)+f(1) = 47 < 55$ for $d = 10$, $f(10)+f(1) = 57 < 66$ for $d = 11$, $f(11)+f(1) = 68 < 78$ for $d = 12$, and $f(7)+f(2) = 33 < 45$ for $d = 9$.) Two single points $p \ne q$ force $\dim \operatorname{aff} X \ge \sum a_i + 1$ by (iv), which excludes $(9,0,0)$, $(8,1,0,0)$ and everything with $\sum a_i = 9$ and two single points; three single points p, q, r would have to lie, by (iv), on a line orthogonal to U_1 when $\sum a_i = 8$, but three collinear points cannot be pairwise at distance 2, which excludes $(8,0,0,0)$. In $(8,1,0)$ the centre c_2 of the pair X_2 is its midpoint, at distance $1/\sqrt{2}$ from the points of X_2, whereas the isolated point is at distance 2 from them; so $c_2 \ne c_3$ and (iv) gives $\dim \operatorname{aff} X \ge 10$, a contradiction. This leaves the listed cases. In (a1) and (b), (iv) forces $c_1 = p$, so $\rho_1^2 = 4 - \|c_1 - p\|^2 = 4$. In (a2), $\rho_1^2 = 4 - \|c_1 - p\|^2 \le 4$. In (a3), (iv) forces $c_1 = c_2$; since $\Gamma(X_2)$ is connected, X_2 is a pair at squared distance 2, so $\rho_2^2 = 1/2$ and (iii) gives $\rho_1^2 = 7/2$. In (a4), c_1, p, q lie on a line orthogonal to U_1 (the orthogonal complement of U_1 in $\operatorname{aff} X$ is one-dimensional by (iv)), $\|p - q\| = 2$, and $\|c_1 - p\| = \|c_1 - q\| = \sqrt{4 - \rho_1^2}$ by (iii); so c_1 is the midpoint of p, q and $\rho_1^2 = 4 - 1 = 3$. □

Note that in every case of Corollary 4.2 the large component X_1 is itself a *connected* configuration, hence (Lemma 3.3) of the form $p + (\{0\} \cup C)$ with C a clique of $G(R)$ for a root lattice R of rank a_1. Sphericity and circumradius of $\{0\} \cup C$ are decided exactly: $\{0\} \cup C$ lies on a sphere with centre c iff $\langle c, v\rangle = \|v\|^2/2$ for all $v \in C$, a linear system over $\mathbb{Q}$; if it is consistent, the least-norm solution c_0 lies in $\operatorname{span} C$ and $\rho^2 = \|c_0\|^2$.

5. Computation

5.1. **Symmetry-reduced clique search.** For each root lattice R we construct $G(R)$ from the coordinates in Section 2 (all vectors scaled by 2 so that coordinates are integers). The largest graphs are $G(E_8 \oplus A_1)$ with 2882 vertices in rank 9, and $G(D_{12})$, $G(E_8 \oplus D_4)$ with 8208 vertices in rank 12.

The graphs $G(R)$ are dense and highly symmetric, and the cliques of interest are large (44 in rank 9). A plain branch-and-bound with greedy colouring bounds (in the style of [16]) does not terminate in reasonable time on $G(E_8)$. We therefore use *orbital branching*, whose correctness rests on the following elementary observation.

Lemma 5.1. *Let G be a graph, S a clique of G, $P \subset V(G) \setminus S$ a set of vertices adjacent to every vertex of S, and $H \le \operatorname{Aut}(G)$ a group fixing S pointwise and mapping P onto itself. Let $v \in P$ and $O = Hv$. Then every clique K with $S \subset K \subset S \cup P$ is the image under an element of H of a clique K' with either $S \cup \{v\} \subset K' \subset S \cup \{v\} \cup (P \cap N(v))$ or $S \subset K' \subset S \cup (P \setminus O)$.*

Proof. If $K \cap O = \emptyset$ take $K' = K$. Otherwise K contains hv for some $h \in H$, and $K' = h^{-1}K$ is a clique containing $S \cup \{v\}$ and contained in $S \cup P$; all its elements other than those of $S \cup \{v\}$ are adjacent to v. □

We apply this with $H = H_S := \langle s_r : r \in R_2,\ \langle r, x\rangle = 0\ \forall x \in S\rangle$. Each such s_r is an automorphism of $G(R)$ and fixes S pointwise, and H_S preserves the current candidate set P (the common neighbourhood of S intersected with the set of not-yet-excluded vertices, which is H_S-invariant by induction). By a theorem of Steinberg [14] (see [9, Thm. 1.12]), H_S is in fact the full pointwise stabiliser of S in $W(R)$, so nothing is lost inside $W(R)$; but only the trivial inclusion $H_S \subseteq \operatorname{Aut}(G(R))_S$ is needed for correctness. The search is the standard one: at a node (S, P) compute a greedy colouring of $G[P]$ and prune if $|S| + \chi_{\text{greedy}}(P)$ cannot exceed the

best clique found (or the target size); otherwise pick the vertex $v \in P$ of highest colour, recurse on $(S \cup \{v\}, P \cap N(v))$, compute the orbit $O = H_S v$ by breadth-first search over the generating reflections, delete O from P and repeat. Orbits are computed only while H_S is nontrivial (i.e. while some root is orthogonal to S); afterwards the plain algorithm is used. The same routine with the pruning rule "$|S| + \chi_{\text{greedy}}(P) < T$" enumerates, for every clique of size $\geq T$, at least one $W(R)$-image (Lemma 5.1 applied inductively), which is what the classification needs. The implementation is a few hundred lines of C using bitsets.

5.2. **Results.** Table 1 lists $f(a)$ for $a \leq 12$ and the lattices attaining it; Table 2 in the appendix gives $\omega(R) + 1$ for all 57 root lattices of rank 9. The total number of search nodes over all 566 root lattices of rank ≤ 12 is 1.5×10^7, and the total solver time is about 70 seconds on one core (all but 65 seconds of it for rank 12; rank 9 takes 0.15 seconds).

a	$f(a)$	lattices R of rank a with $1 + \omega(R) = f(a)$
1	2	A_1
2	4	$A_1 \oplus A_1$ (the square)
3	6	A_3 (the octahedron)
4	10	A_4
5	16	D_5
6	27	E_6
7	29	E_7
8	45	E_8
9	45	A_9, $E_8 \oplus A_1$ (next: D_9 with 37, $A_8 \oplus A_1$ with 36)
10	55	A_{10} (next: D_{10} with 46; $E_8 \oplus A_2$, $E_8 \oplus A_1^2$, $A_9 \oplus A_1$ with 45)
11	66	A_{11} (next: D_{11} with 56, $A_{10} \oplus A_1$ with 55)
12	78	A_{12} (next: D_{12} with 67, $A_{11} \oplus A_1$ with 66)

TABLE 1. The maximum size $f(a)$ of a connected configuration with squared distances $2, 4$ in $\mathbb{R}^a$, over all root lattices of rank $\leq a$ ($39, 57, 88, 128, 193$ lattices for $a = 8, \ldots, 12$).

For $a \leq 8$ the values agree with $g(a)$ for $a \geq 3$ (Section 7); note $f(2) = 4 < g(2) = 5$ since the regular pentagon has $|X| = 5 = 2d + 1$, outside the range of Theorem 2.1, and likewise $f(1) = 2 < g(1) = 3$. The maximum in rank 9 is attained exactly twice: by A_9, where the 44-clique together with the origin is $J(10, 2)$ (the vectors $e_i + e_j - e_9 - e_{10}$, $1 \leq i < j \leq 10$, in $A_9 \subset \mathbb{Z}^{10}$, translated), and by $E_8 \oplus A_1$, where every maximum clique lies in E_8 and gives L_8. The sequence $1 + \omega(D_a) = 1 + \binom{a}{2}$ for $8 \leq a \leq 12$ is realised by $\{0\} \cup \{e_i + e_j\}$.

5.3. **Near-maximum cliques and sphericity.** The enumeration routine was run to produce all cliques of size $\geq T$ up to $W(R)$ in the lattices relevant to Corollary 4.2. Distinct outputs were sorted into similarity classes by the isomorphism type of the short-distance graph (which determines the Gram matrix, hence the configuration up to isometry), and each class was tested for sphericity by exact rational linear algebra. The results are:

- E_8, $T = 44$: one class, the 45-point set L_8; its short-distance graph has degrees 36 (nine vertices, a regular simplex) and 21 (thirty-six vertices, $J(9, 2)$); non-spherical.
- E_8, $T = 43$: two classes of 44-point sets, namely L_8 minus a point of either type; both non-spherical.
- E_8, $T = 42$: 10 861 output cliques of size 42 (43-point sets), *all non-spherical*; in particular none has $\rho^2 \in \{3, 7/2\}$.

- $E_8 \oplus A_1$, $T = 43$: exactly the same three similarity classes as for E_8 (all outputs have affine dimension 8); in particular $G(E_8 \oplus A_1)$ has no 44-clique other than those giving L_8.
- A_9, $T = 43$: one class of 45-point sets ($J(10,2)$, spherical, $\rho^2 = 8/5$) and one class of 44-point sets ($J(10,2)$ minus a point, spherical, $\rho^2 = 8/5$).
- Every other root lattice of rank ≤ 9 has $\omega \le 36$.
- A_d for $d = 10, 11, 12$, $T = \binom{d+1}{2} - 2$: exactly two classes, $J(d+1,2)$ and $J(d+1,2)$ minus a point, both spherical with $\rho^2 = 2 - \frac{4}{d+1} \ne 4$. Every other root lattice of rank d has $\omega \le \binom{d}{2}$, so it contains no clique of size $\binom{d+1}{2} - 2 \ge 53$.

6. Proofs of the theorems

Proof of Theorem 1.1. Let $X \subset \mathbb{R}^9$ be a two-distance set with $|X| \ge 45 > 2\cdot 9+1$; by Theorem 2.1 we may assume the squared distances are 2 and 4. If $\Gamma(X)$ is connected, then by Lemma 3.3 $|X| \le f(9) = 45$, and if $|X| = 45$ then X is, up to translation, $\{0\} \cup C$ for a 44-clique C of $G(R)$ with R of rank 9; by Table 2 $R \in \{A_9, E_8 \oplus A_1\}$, and by Section 5.3 X is $J(10,2)$ or L_8. If $\Gamma(X)$ is disconnected, Corollary 4.2(a) applies with $|X| \ge 45$. In case (a1) the large component is a connected spherical set of radius 2 with at least 44 points spanning $\mathbb{R}^9$; by Section 5.3 the only connected sets with ≥ 44 points in $\mathbb{R}^9$ are L_8 and L_8 minus a point (spanning only an 8-space, and non-spherical), and $J(10,2)$ and $J(10,2)$ minus a point (spherical with $\rho^2 = 8/5 \ne 4$): impossible. In case (a2) the large component is a connected spherical set with ≥ 44 points in an 8-space; the only connected sets with ≥ 44 points in $\mathbb{R}^8$ are L_8 and L_8 minus a point, all non-spherical: impossible. In cases (a3), (a4) the large component is a connected spherical set with ≥ 43 points in an 8-space and $\rho^2 \in \{7/2, 3\}$; by Section 5.3 no connected 43-point set in $\mathbb{R}^8$ is spherical, and the 44- and 45-point ones are not either: impossible. Hence $|X| \le 45$, with equality only for $J(10,2)$ and L_8, both of which are 45-point two-distance sets in $\mathbb{R}^9$. □

Proof of Theorem 1.2. Let $d \in \{10, 11, 12\}$, $N = \binom{d+1}{2}$, and let $X \subset \mathbb{R}^d$ be a two-distance set with $|X| \ge N > 2d+1$, scaled to squared distances $2, 4$. If $\Gamma(X)$ is connected, $|X| \le f(d) = N$ with equality only for $\{0\} \cup C$, C a maximum clique of $G(A_d)$, i.e. $X = J(d+1,2)$. If $\Gamma(X)$ is disconnected, Corollary 4.2(b) requires a connected spherical set of radius 2 with $\ge N-1$ points spanning $\mathbb{R}^d$; the only connected sets with $\ge N-1$ points in $\mathbb{R}^d$ are $J(d+1,2)$ and $J(d+1,2)$ minus a point, with $\rho^2 = 2 - 4/(d+1) \ne 4$. Hence $g(d) = N$ and $J(d+1,2)$ is the unique extremal set. □

Proof of Theorem 1.3. Let $X \subset \mathbb{R}^8$ be a 45-point two-distance set, scaled as usual ($45 > 17$). If $\Gamma(X)$ is connected then $X = \{0\} \cup C$ with C a 44-clique of $G(R)$, R of rank ≤ 8, hence $R = E_8$ and $X = L_8$. If $\Gamma(X)$ is disconnected, then $\sum a_i \le 8$ and $\sum f(a_i) \ge 45$ force $(a_1, a_2) = (8, 0)$ with $c_1 = p$ (Lemma 4.1(iv)), so the large component is a connected spherical 44-point set in $\mathbb{R}^8$, which does not exist by Section 5.3. □

Proof of Theorem 1.4. Theorem 2.1 gives the scaling. For each component X_i with $|X_i| \ge 2$, Lemmas 3.2 and 3.3 applied to X_i give a root lattice $R_i = R(X_i)$ of rank $\le \dim \operatorname{aff}(X_i)$ with $X_i - X_i \subset R_i$; for a single point take $R_i = 0$. The R_i span the mutually orthogonal spaces U_i of Lemma 4.1(i), and (ii) gives sphericity of the components when there are at least two. □

7. Verification

Since the results depend on computation, we describe what was done to guard against errors in the reduction and in the code.

Known values. The pipeline was first run for all root lattices of rank ≤ 8. It returns $f(3), \dots, f(8) = 6, 10, 16, 27, 29, 45$, i.e. exactly $g(3), \dots, g(8)$, attained in A_3 (octahedron), A_4 ($J(5,2)$), D_5 (the 16 vertices of the 5-demicube, degrees 10, $\rho^2 = 5/4$), E_6 (the 27-point Schläfli set, degrees 16, spherical with $\rho^2 = 4/3$), E_7 (a 29-point non-spherical set, degrees $7, 15^{21}, 22^7$) and E_8 (L_8), in agreement with Lisoněk's tables. Enumerating the maximum cliques shows that the 27-, 29- and 45-point sets are unique within their lattices, consistent with [12]. Lattice vector counts agree with the theta series of A_n, D_n, E_6, E_7, E_8.

Two solvers. A plain branch-and-bound without any symmetry reduction (an independent, simpler program) was run in decision mode ("is there a clique of size ≥ 45?") on all 57 lattices of rank 9; it terminates on 56 of them within seconds and confirms the answer "no" in each case (and confirms $\omega(A_9) = 44$, $\omega(D_9) = 36$ exactly). The remaining lattice is $E_8 \oplus A_1$. For E_8 itself the inequality $\omega(E_8) \leq 44$ is a theorem (Blokhuis' bound $g(8) \leq 45$), so $E_8 \oplus A_1$ is the only genuinely new negative case. For it we wrote a third, independent implementation in Python: the reflections s_r were recomputed and each was checked to be an automorphism of $G(E_8 \oplus A_1)$; orbital branching was carried out to depth 4 in Python (orbits by breadth-first search); and each of the resulting 119 subproblems "is there a clique of size $45 - |S|$ in $G[P]$?" was given to a SAT solver (CaDiCaL/Glucose via PySAT, with a sequential-counter cardinality encoding). Subproblems on which the SAT solver exceeded 10 seconds were split one further orbital level, and two of them were instead settled by the plain solver. In total 2299 subproblems were solved, all unsatisfiable, confirming $\omega(E_8 \oplus A_1) \leq 44$. The same symmetry-free check confirms $\omega(A_d) = \binom{d+1}{2} - 1$ and $\omega(A_{d-1} \oplus A_1) \leq \binom{d+1}{2} - 1$ for $d = 10, 11, 12$ and $\omega(D_{10}) \leq 54$; the remaining rank 10–12 lattices rest on the orbital solver alone.

Certificates. A standalone script (`verify.py`, standard library only, exact rational arithmetic) checks that $J(10,2)$, $J(11,2)$, $J(12,2)$, $J(13,2)$ and L_8 are two-distance sets of the stated sizes and dimensions with the stated circumradii, that L_8 is not spherical, that no 44-point subset of L_8 is spherical and that no 43-point subset of L_8 is spherical with $\rho^2 \in \{3, 7/2\}$, that the lattice vector counts are correct, and that the clique numbers of $G(A_3)$, $G(D_4)$, $G(A_4)$ are $5, 7, 9$ by brute force.

What is proved and what is computed. Theorems 2.1, Lemmas 3.1–4.1 and Corollary 4.2 (given the values of f) are proved by hand. The values $f(a)$, $a \leq 12$, the clique numbers in Table 2, and the enumerations of Section 5.3 are the results of exhaustive computer searches as described; their correctness relies on Lemma 5.1 and on the implementations, which were cross-validated as above.

8. Remarks

Remark 8.1. Theorem 2.1 allows $k = 3$ (ratio $\sqrt{3/2}$) from $d = 13$ on. The spectral bounds of Chen and Yu [4, Table 2] show that for $13 \leq d \leq 21$ every two-distance set with $k \geq 3$ has fewer than $\binom{d+1}{2}$ points (their bound for $k = 3$ is 177 at $d = 21$ but 277 at $d = 22$). Consequently, for all $d \leq 21$, deciding whether $g(d) = \binom{d+1}{2}$ (and classifying the extremal sets) is a finite computation of exactly the kind performed here, on the graphs $G(R)$ of root lattices of rank d. The graphs grow quickly ($|R_4| = 2d + 16\binom{d}{4}$ for D_d), so new ideas will be needed well before $d = 21$. (Dimension 23 is different: the 277-point set of [17] has $k = 3$.)

Remark 8.2. The appearance of E_8 in dimension 8 and its disappearance from $d = 9$ on has a simple explanation in this framework: the 45-point set is a 44-clique of $G(E_8)$, and adjoining orthogonal summands to E_8 does not create larger cliques ($\omega(E_8 \oplus A_1) = \omega(E_8 \oplus A_2) = \omega(E_8 \oplus A_1^2) = 44$, $\omega(E_8 \oplus A_3) = 44$, $\omega(E_8 \oplus D_4) = 44$), while $\omega(A_d) = \binom{d+1}{2} - 1$ grows quadratically.

Remark 8.3. The second-largest connected configurations in ranks 8–12 come from D_a and are the sets $\{0\} \cup \{e_i + e_j\}$ of size $1 + \binom{a}{2}$. It would be interesting to know whether the D_a family or something else provides the largest non-Johnson two-distance sets in higher dimensions.

Data and code

The accompanying files are: `lattices.py` (lattice vectors and the graphs $G(R)$), `orb.c` and `orbdrive.py` (orbital branch-and-bound, checkpointed driver), `sweep.py` (all root lattices of a given rank), `analyze.py` (enumeration, similarity classes, exact sphericity test), `mcs.c` (plain solver), `referee.py` (independent SAT-based check), `verify.py` (standalone certificate check), and the JSON result files (`jobs/sweep_rank*.json`) containing $\omega(R)$ and a maximum clique for every root lattice of rank ≤ 12.

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Appendix A. The 57 root lattices of rank 9

R	$\lvert V(G(R))\rvert$	$1+\omega$	R	$\lvert V(G(R))\rvert$	$1+\omega$
E_8A_1	2882	45	$A_6A_1^3$	522	21
A_9	1350	45	$A_2A_6A_1$	608	21
D_9	2178	37	$A_4A_3A_1^2$	444	20
A_8A_1	974	36	$D_5A_3A_1$	734	20
$E_7A_1^2$	1394	30	$D_4A_4A_1$	668	20
D_8A_1	1474	30	$A_4A_3A_2$	506	20
E_7A_2	1644	30	D_5D_4	1138	20
A_5A_4	770	30	$D_6A_1^3$	690	19
$A_7A_1^2$	708	28	$D_6A_2A_1$	812	19
A_7A_2	818	28	$A_5A_2A_1^2$	458	18
A_6A_3	774	28	$A_5A_2^2$	528	18
$E_6A_1^3$	792	27	$D_5A_1^4$	482	16
$E_6A_2A_1$	938	27	$A_3^2A_1^3$	342	16
E_6A_3	1224	27	$D_5A_2A_1^2$	568	16
$A_4^2A_1$	582	25	$D_4A_3A_1^2$	506	16
D_5A_4	980	25	$A_3^2A_2A_1$	392	16
$D_7A_1^2$	1002	24	$D_4^2A_1$	770	16
$A_5A_3A_1$	584	24	$D_5A_2^2$	658	16
D_7A_2	1168	24	$D_4A_3A_2$	576	16
D_6A_3	1050	24	A_3^3	486	16
D_4A_5	888	24	$A_5A_1^4$	392	15
$A_4A_2A_1^3$	350	15	$A_4A_2^2A_1$	404	15
$A_4A_1^5$	300	12	$D_4A_1^5$	338	12
$A_3A_2A_1^4$	272	12	$D_4A_2A_1^3$	396	12
$A_3A_2^2A_1^2$	314	12	$D_4A_2^2A_1$	458	12
$A_3A_2^3$	360	12	$A_3A_1^6$	234	11
A_1^9	162	10	$A_2A_1^7$	188	10
$A_2^2A_1^5$	218	10	$A_2^3A_1^3$	252	10
$A_2^4A_1$	290	10			

TABLE 2. Number of vertices of $G(R)$ and $1+\omega(R)$ for every root lattice R of rank 9 ($\oplus$ signs omitted).

The authors are listed in alphabetical order and contributed equally.

(B. Z. Bu) University of Massachusetts Amherst

(K. Gill) Stanford Online High School